

OpenStax Calculus Volume 1

Section 4.2 Exercises: Linear Approximations and Differentials

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1. What is the linear approximation for any generic linear function $y = mx + b$?
2. Determine the necessary conditions such that the linear approximation function is constant. Use a graph to prove your result.
3. Explain why the linear approximation becomes less accurate as you increase the distance between x and a . Use a graph to prove your argument.
4. When is the linear approximation exact?

For the following exercises, find the linear approximation $L(x)$ to $y = f(x)$ near $x = a$ for the function.

5. $f(x) = x + x^4, a = 0$
6. $f(x) = \frac{1}{x}, a = 2$
7. $f(x) = \tan x, a = \frac{\pi}{4}$
8. $f(x) = \sin x, a = \frac{\pi}{2}$
9. $f(x) = x \sin x, a = 2\pi$
10. $f(x) = \sin^2 x, a = 0$

For the following exercises, compute the values given within 0.01 by deciding on the appropriate $f(x)$ and a , and evaluating $L(x) = f(a) + f'(a)(x - a)$. Check your answer using a calculator.

11. [T] $(2.001)^6$
12. [T] $\sin(0.02)$
13. [T] $\cos(0.03)$
14. [T] $(15.99)^{1/4}$
15. [T] $\frac{1}{0.98}$
16. [T] $\sin(3.14)$

For the following exercises, determine the appropriate $f(x)$ and a , and evaluate $L(x) = f(a) + f'(a)(x - a)$. Calculate the numerical error in the linear approximations that follow.

17. [T] $(1.01)^3$
18. [T] $\cos(0.01)$
19. [T] $(\sin(0.01))^2$
20. [T] $(1.01)^{-3}$
21. [T] $(1 + \frac{1}{10})^{10}$
22. [T] $\sqrt{8.99}$

For the following exercises, find the differential of the function.

23. $y = 3x^4 + x^2 - 2x + 1$
24. $y = x \cos x$
25. $y = \sqrt{1+x}$
26. $y = \frac{x^2+2}{x-1}$

For the following exercises, find the differential and evaluate for the given x and dx .

27. $y = 3x^2 - x + 6$, $x = 2$, $dx = 0.1$
28. $y = \frac{1}{x+1}$, $x = 1$, $dx = 0.25$
29. $y = \tan x$, $x = 0$, $dx = \frac{\pi}{10}$
30. $y = \frac{3x^2+2}{\sqrt{x+1}}$, $x = 0$, $dx = 0.1$
31. $y = \frac{\sin(2x)}{x}$, $x = \pi$, $dx = 0.25$
32. $y = x^3 + 2x + \frac{1}{x}$, $x = 1$, $dx = 0.05$

For the following exercises, find the change in volume dV or in surface area dA .

33. dV if the sides of a cube change from 10 to 10.1.
34. dA if the sides of a cube change from x to $x + dx$.
35. dA if the radius of a sphere changes from r by dr .
36. dV if the radius of a sphere changes from r by dr .
37. dV if a circular cylinder with $r = 2$ changes height from 3 cm to 3.05 cm.
38. dV if a circular cylinder of height 3 changes from $r = 2$ to $r = 1.9$ cm.
39. A spherical golf ball is measured to have a radius of 5 mm, with a possible measurement error of 0.1 mm. What is the possible change in volume?
40. A pool has a rectangular base of 10 ft by 20 ft and a depth of 6 ft. What is the change in volume if you only fill it up to 5.5 ft?

41. An ice cream cone has height 4 in. and radius 1 in. If the cone is 0.1 in. thick, what is the difference between the volume of the cone, including the shell, and the volume of the ice cream you can fit inside the shell?

For the following exercises, confirm the approximations by using the linear approximation at $x = 0$.

42. $\sqrt{1-x} \approx 1 - \frac{1}{2}x$

43. $\frac{1}{\sqrt{1-x^2}} \approx 1$

44. $\sqrt{c^2 + x^2} \approx c$